

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

MERN STACK

Practical – VII

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Class: TYIT	Batch:1
Date of Assignment: 19-09-26	Date/Time of Submission: 19-09-26

7.MongoDB Basics

a. Install and start MongoDB.

Step 1: Download MongoDB

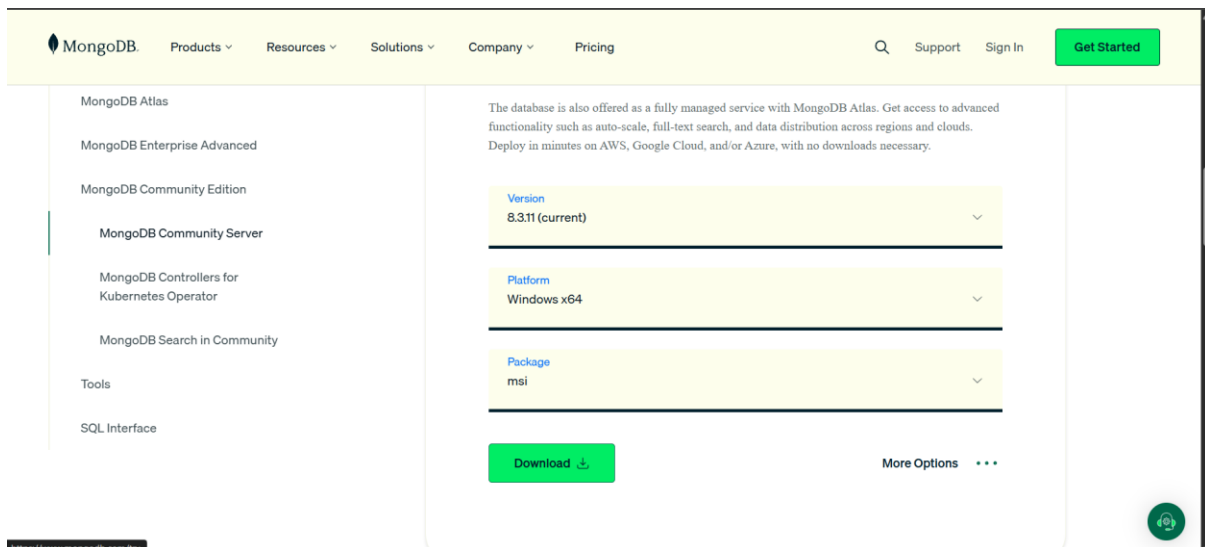
Go to the official MongoDB website and download MongoDB Community Server for Windows:

MongoDB Community Server Download

Choose:

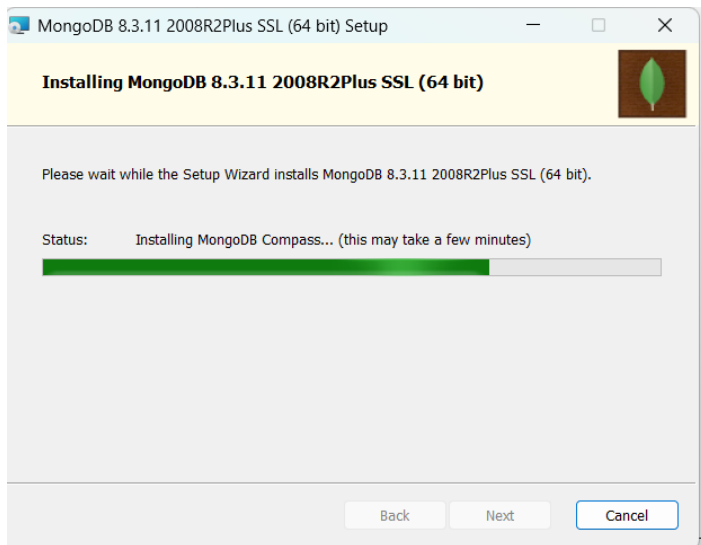
- Version: Current/available version
- Platform: Windows
- Package: MSI

Download the .msi installer.

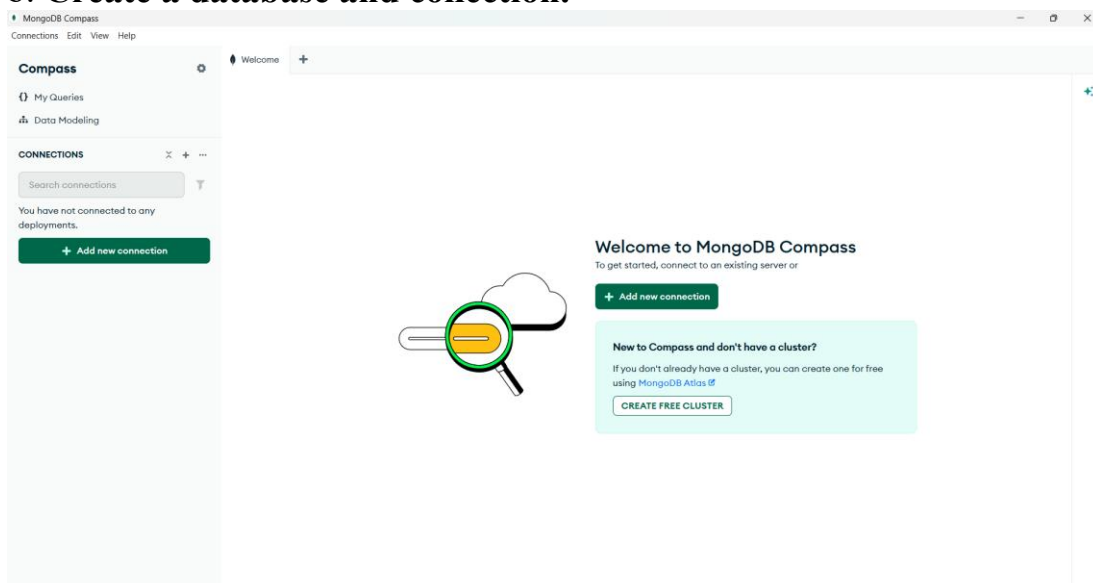


Step 2: Install MongoDB

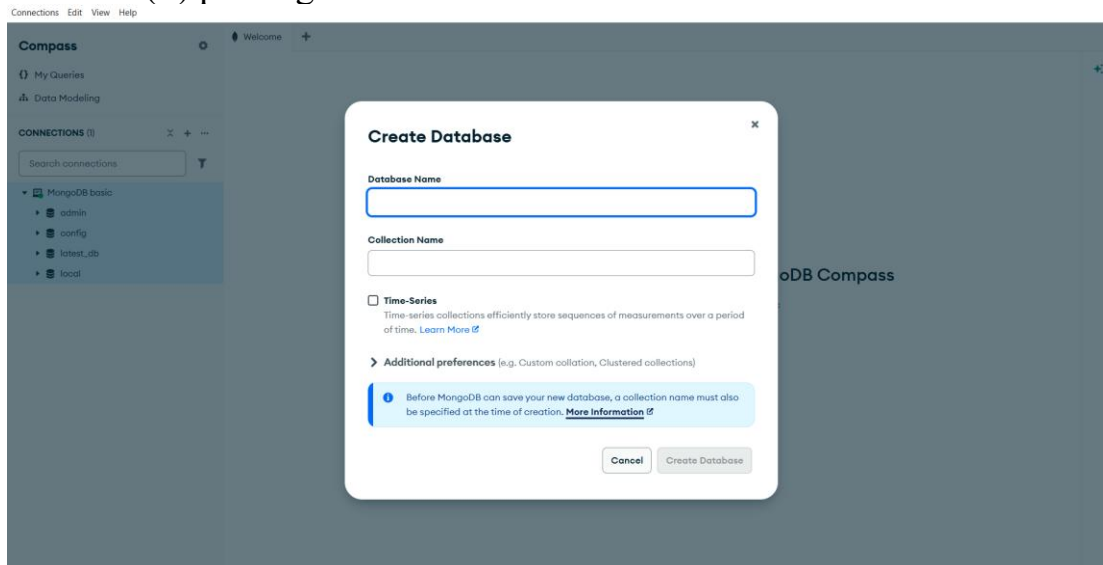
1. Open the downloaded .msi file.
2. Click Next.
3. Accept the license agreement.
4. Select Complete installation.
5. Keep the default installation settings.
6. Click Install.
7. After installation, click Finish.
8. You will see following window



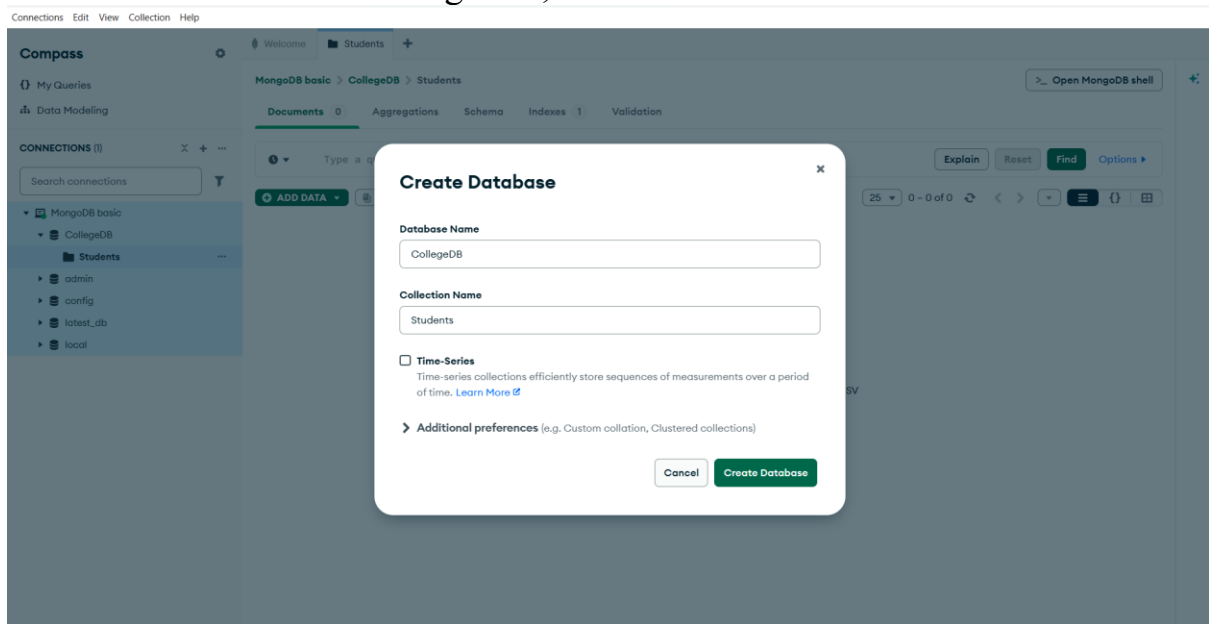
b. Create a database and collection.



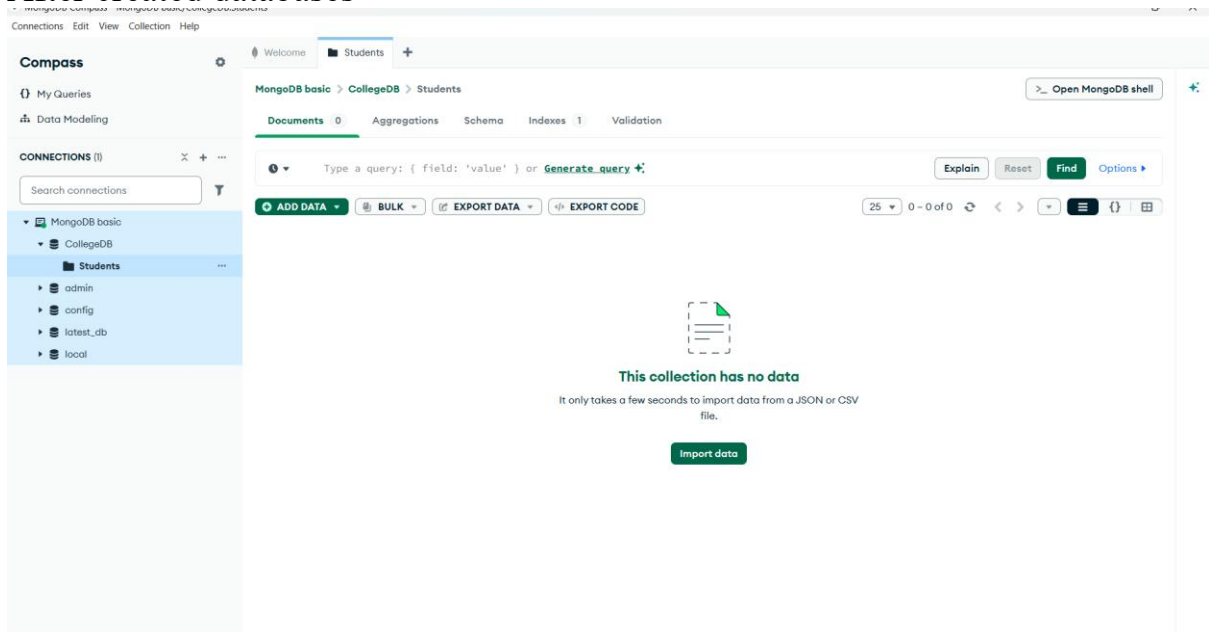
Click on (+) plus sign and create database.



Enter Database Name - CollegeDB , collection name – Students

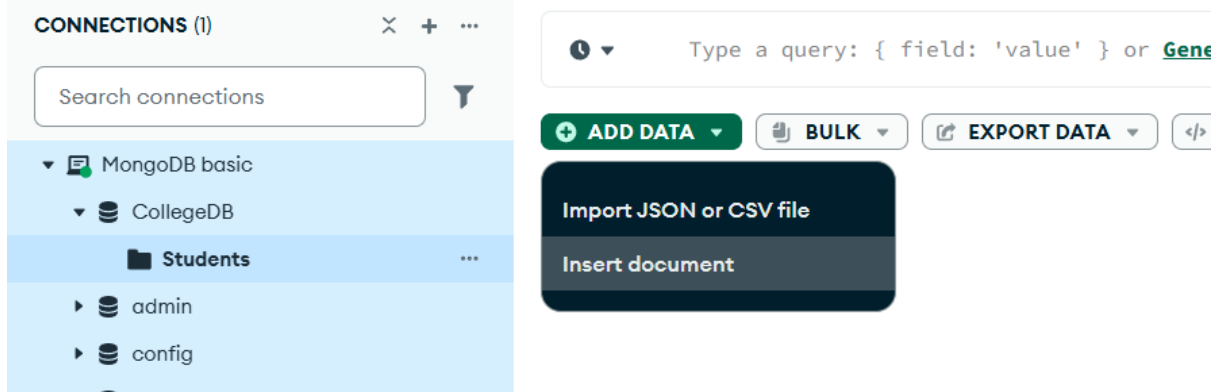


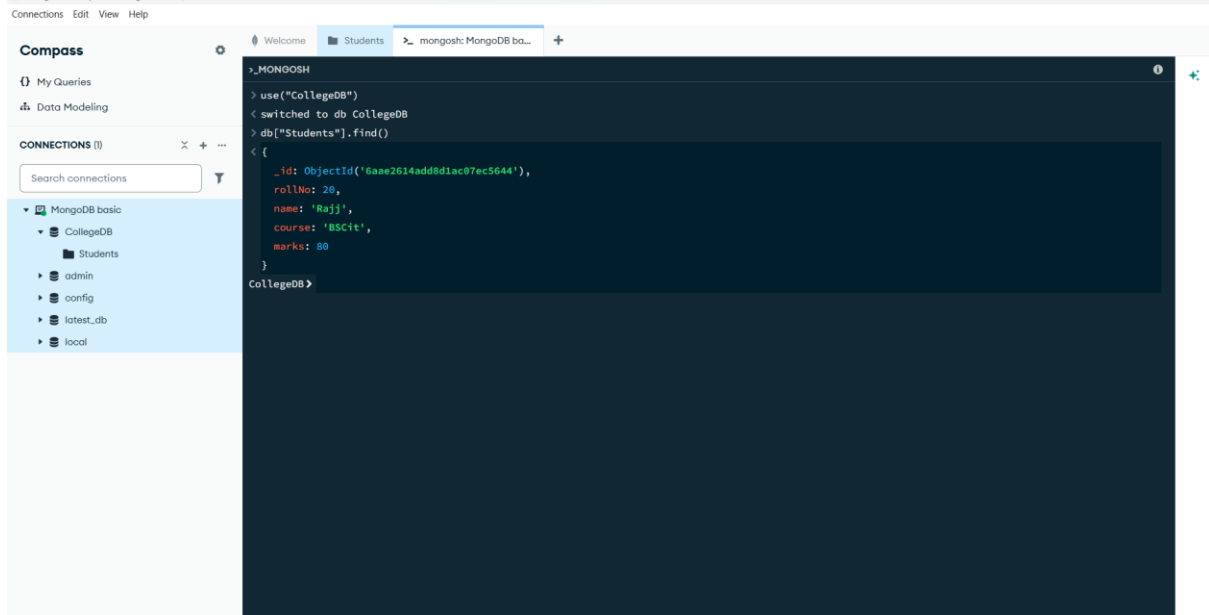
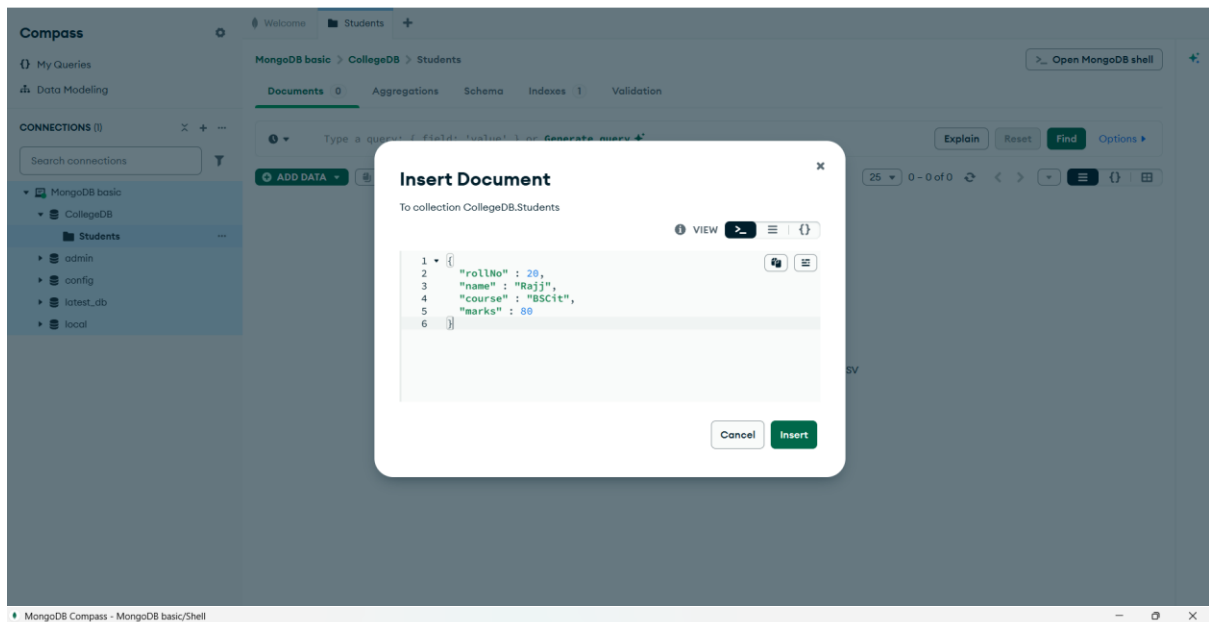
After created databases



c. Insert records in MongoDB.

Now click ADD DATA -> Insert document





Inserting single recorde

```
> db.Students.insertOne({
  rollNo : 102,
  name : "ashwin",
  course : "BSCIT",
  marks : 70
})
```

After inserting you will get acknowledgement

```
< {
  acknowledged: true,
  insertedId: ObjectId('6aae279a1f2057f30a980562')
}
```

Inserting multiple recorde

```

> db.Students.insertMany([
  {
    rollNo : 103,
    name : "pratik",
    course : "BCA",
    marks : 75
  },
  {
    rollNo : 104,
    name : "khushi",
    course : "BSCIT",
    marks : 80
  },
  {
    rollNo : 105,
    name : "sachin",
    course : "BSCA",
    marks : 78
  }
] )
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6aae29b41f2057f30a980563'),
    '1': ObjectId('6aae29b41f2057f30a980564'),
    '2': ObjectId('6aae29b41f2057f30a980565')
  }
}

```

d. View records from the collection.

db.students.find() - View all records. This displays all student documents.

db.students.find().pretty() - View records in a readable format

```

< {
  _id: ObjectId('6aae2614add8d1ac07ec5644'),
  rollNo: 20,
  name: 'Rajj',
  course: 'BSCit',
  marks: 80
}
{
  _id: ObjectId('6aae27081f2057f30a980560')
}
{
  _id: ObjectId('6aae27331f2057f30a980561')
}
{
  _id: ObjectId('6aae279a1f2057f30a980562'),
  rollNo: 102,
  name: 'ashwin',
  course: 'BSCIT',
  marks: 70
}
{
  _id: ObjectId('6aae29b41f2057f30a980563'),
  rollNo: 103,
  name: 'pratik',
  course: 'BCA',
  marks: 75
}
{
  _id: ObjectId('6aae29b41f2057f30a980564'),

```

Find with roll no :

```

> db.Students.find({rollNo : 105})
< {
  _id: ObjectId('6aae29b41f2057f30a980565'),
  rollNo: 105,
  name: 'sachin',
  course: 'BSCA',
  marks: 78
}

```

e. Update and delete records.

Delete - Delete the student with roll number 104.

deleteOne - deletes only 1 document

deleteMany - deletes all matching documents

Delete one data

```

> db.Students.deleteOne({rollNo:105})
< {
  acknowledged: true,
  deletedCount: 1
}
> db.Students.find({rollNo : 105})
<

```

Update many ..

```

> db.Students.find({rollNo : 105})
<
> db.Students.updateMany(
  {
    course : "BSCIT"
  },
  {$set : {course : "Information technology"}}
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
}
{
  _id: ObjectId('6aae27081f2057f30a980560')
}
{
  _id: ObjectId('6aae27331f2057f30a980561')
}
{
  _id: ObjectId('6aae279a1f2057f30a980562'),
  rollNo: 102,
  name: 'ashwin',
  course: 'Information technology',
  marks: 70
}
{
  _id: ObjectId('6aae29b41f2057f30a980563'),
  rollNo: 103,
  name: 'pratik',
  course: 'BCA',
  marks: 75
}
{
  _id: ObjectId('6aae29b41f2057f30a980564'),
  rollNo: 104,
  name: 'khushi',
  course: 'Information technology',
  marks: 80
}
}

```

f. Connect MongoDB with Node.js.

Code :connecting mongodb with node server.

```
const mongoose = require('mongoose');

mongoose.connect("mongodb://localhost:27017/CollegeDB")
.then(() => {
  console.log("Connection Successful");
})
.catch((err) => {
  console.log("Connection Failed");
  console.log(err);
});
```

Output :

```
PS R:\MERN STACK\practical7> node server.js
Connection Successful
```

Code :Fetching data from the database

```
PS R:\MERN STACK\practical7> node --watch server.js
Connection Successful
[
  {
    _id: new ObjectId('6aae2614add8d1ac07ec5644'),
    rollNo: 20,
    name: 'Rajj',
    course: 'BSCit',
    marks: 80
  },
  {
    _id: new ObjectId('6aae27081f2057f30a980560'),
    _id: new ObjectId('6aae27331f2057f30a980561'),
    _id: new ObjectId('6aae279a1f2057f30a980562'),
    rollNo: 102,
    name: 'ashwin',
    course: 'Information technology',
    marks: 70
  },
  {
    _id: new ObjectId('6aae29b41f2057f30a980563'),
    rollNo: 103,
    name: 'pratik',
    course: 'BCA',
    marks: 75
  },
  {

```

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